

PREDICTIVE RISK INTELLIGENCE FOR METABOLIC SCREENING IN DIABETES

A Semi-Supervised Deep Learning Framework for Early Detection of

Metabolic Dysfunction in Normal-BMI Adults

A Dissertation submitted in partial fulfilment of the requirements

for the degree of Master of Computer Applications

Submitted by

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June 2026

CERTIFICATE

This is to certify that the dissertation entitled

“Predictive Risk Intelligence for Metabolic Screening in Diabetes”

submitted by _____ (Registration No.

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degree of Master of Computer Applications has been carried out under my supervision.

The work reported in this dissertation is original and has not been submitted elsewhere for any degree.

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DECLARATION

We hereby declare that the work presented in this dissertation, titled **“Predictive Risk Intelligence for Metabolic Screening in Diabetes”**, submitted to _____ for the award of the degree of Master of Computer Applications, is our collective original work. The dissertation has not been submitted for any other degree or examination in this or any other university.

All sources of information used in this work have been duly cited and referenced. Any ideas or data taken from other sources have been acknowledged according to standard referencing practices.

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ACKNOWLEDGEMENTS

We would like to express our sincere gratitude to our supervisor for their guidance, patience, and constructive feedback throughout this project. Their advice helped us approach the clinical and machine learning aspects of this problem with honesty and rigour.

We are grateful to our department and university for providing the computational resources and academic environment necessary to pursue this research. We also thank our families and friends for their constant support.

This work makes use of publicly available data from the National Health and Nutrition Examination Survey (NHANES), provided by the Centers for Disease Control and Prevention (CDC). We thank the survey participants whose anonymised data made this research possible.

ABSTRACT

Body Mass Index (BMI) is the predominant screening metric in primary care cardiometabolic assessment. However, in adults with normal BMI (18.5–24.9 kg/m²), this proxy offers limited discriminative utility regarding fat distribution, visceral adiposity, and ectopic lipid accumulation. Some normal-weight individuals may exhibit concurrent insulin resistance and hepatic steatosis without triggering standard screening thresholds—a combination associated with increased cardiovascular and hepatic disease risk.

This dissertation, Predictive Risk Intelligence for Metabolic Screening in Diabetes, presents the Latent Metabolic State Inference System (LMSIS), a semi-supervised deep learning framework for early detection of metabolic dysfunction in normal-BMI adults. The core model, the **Dual-Anchored Semi-Supervised Identifiable Variational Autoencoder (DA-SS-iVAE)**, is designed according to Nonlinear ICA identifiability principles (up to coordinate permutation and element-wise monotone scaling) through demographic-conditioned priors and regularises biological interpretability through monotone anchor networks aligned to reference-standard measurements: HOMA-IR for insulin resistance (Z_1) and FibroScan Controlled Attenuation Parameter (CAP) for hepatic steatosis (Z_2).

Trained exclusively on the NHANES J-cycle (2017–2018, $n_{\{ \textit{train} \}} = 401$, random 70% of J-cycle complete records), the DA-SS-iVAE was evaluated on the strict temporal holdout P-cycle (2019–2020), which contains zero training participants. A five-seed robustness analysis yields a mean P-cycle Spearman correlation of $|\rho| = 0.583 \pm 0.027$ against FibroScan CAP ($n = 870$; all five seeds exceed the mixed-split

baseline of 0.501). An AUROC of 0.841 is achieved when the latent steatosis coordinate Z_2 is evaluated against FibroScan CAP thresholds ($CAP \geq 248$ dB/m). Mondrian conformal calibration restores $\geq 90\%$ subgroup coverage for the highest-risk Dual-Burden quadrant, which achieves only 81.6% coverage under standard marginal calibration.

All results are reported with their full uncertainty ranges, sample-size limitations, and simulation-induced inflation disclaimers. The system is implemented as a full-stack research prototype comprising a PyTorch inference engine, a FastAPI backend service, and an interactive React-based Research Translation Layer.

Keywords: Identifiable Variational Autoencoder, Conformal Prediction, Metabolic Phenotyping, Normal-BMI MASLD, Semi-Supervised Learning, Symbolic Regression, Riemannian Geometry, Clinical Machine Learning Systems.

LIST OF ABBREVIATIONS

Abbreviation	Expansion
AIP	Atherogenic Index of Plasma
ALT	Alanine Aminotransferase
AST	Aspartate Aminotransferase
AUROC	Area Under the Receiver Operating Characteristic Curve
BMI	Body Mass Index
BUN	Blood Urea Nitrogen
CAP	Controlled Attenuation Parameter
CDC	Centers for Disease Control and Prevention
DA-SS-iVAE	Dual-Anchored Semi-Supervised Identifiable Variational Autoencoder
DCR	Discriminative Contribution Ratio
ELBO	Evidence Lower Bound
FLI	Fatty Liver Index
GGT	Gamma-Glutamyl Transferase
HDL	High-Density Lipoprotein
HOMA-IR	Homeostatic Model Assessment for Insulin Resistance
HSI	Hepatic Steatosis Index

ICA	Independent Component Analysis
IR	Insulin Resistance
iVAE	Identifiable Variational Autoencoder
LDL	Low-Density Lipoprotein
LMSIS	Latent Metabolic State Inference System
MASLD	Metabolic Dysfunction-Associated Steatotic Liver Disease
MASH	Metabolic Dysfunction-Associated Steatohepatitis
MetS	Metabolic Syndrome
MHNW	Metabolically Healthy Normal Weight
MLP	Multi-Layer Perceptron
NAFLD	Non-Alcoholic Fatty Liver Disease
NHANES	National Health and Nutrition Examination Survey
OOD	Out-of-Distribution
PySR	Python Symbolic Regression
TyG	Triglyceride-Glucose Index
VAE	Variational Autoencoder
VCTE	Vibration-Controlled Transient Elastography

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INTRODUCTION

Introduction

Body Mass Index (BMI) is the predominant screening metric in primary care cardiometabolic assessment. While BMI correlates with total adiposity in obese populations, it offers limited discriminative utility regarding regional fat distribution, visceral adiposity, and ectopic lipid accumulation. In adults with normal BMI (18.5–24.9 kg/m²), this proxy may fail to identify biochemically hidden metabolic risk factors: a substantial proportion may exhibit concurrent insulin resistance and hepatic steatosis—conditions associated with increased cardiovascular and hepatic disease risk in the absence of standard screening alerts.

This dissertation, Predictive Risk Intelligence for Metabolic Screening in Diabetes, presents the Latent Metabolic State Inference System (LMSIS), a semi-supervised deep learning framework for early detection of metabolic dysfunction in normal-BMI adults. The core model—the Dual-Anchored Semi-Supervised Identifiable Variational Autoencoder (DA-SS-iVAE)—is designed according to Nonlinear ICA identifiability principles (up to coordinate permutation and element-wise monotone scaling) through demographic-conditioned priors and regularises biological interpretability through monotone anchor networks aligned to reference-standard measurements: HOMA-IR for insulin resistance (Z_1) and FibroScan Controlled Attenuation Parameter (CAP) for hepatic steatosis (Z_2).

The system is implemented as a full-stack research prototype comprising a PyTorch-based inference engine, a FastAPI backend service, and an interactive React-based Research Translation Layer. All components are version-controlled, integration-

tested, and publicly documented. This work is situated at the intersection of deep generative modelling, conformal uncertainty quantification, and clinical machine-learning systems engineering.

Existing Systems

Contemporary metabolic screening relies on two categories of tools: clinical scoring indices and supervised machine-learning classifiers.

Clinical Scoring Indices. The Fatty Liver Index (FLI), Hepatic Steatosis Index (HSI), Triglyceride-Glucose Index (TyG), and NAFLD Liver Fat Score (LFS) are widely used in primary care. These indices combine BMI, liver enzymes, lipid panels, and glucose metrics into scalar risk scores. They are inexpensive, interpretable, and validated in large mixed-BMI cohorts.

Machine Learning Classifiers. Recent work applies deep neural networks and gradient-boosted trees to wearable sensor data [Reference] and NHANES biomarkers [Reference] for MASLD risk estimation. Unsupervised approaches such as the SENA- δ VAE [Reference] model causal discrepancies in genetic perturbation data but do not anchor latent factors to clinically measurable phenotypes.

Among the systems identified in the reviewed literature, all share a critical limitation when restricted to normal-BMI populations: they are calibrated on mixed-BMI cohorts where BMI contributes substantial discriminative variance. When applied to normal-BMI adults, the reduced variance of BMI diminishes the predictive signal of these models, producing degraded or inverted risk rankings.

Problem Statement

Standard cardiometabolic screening workflows use BMI as a primary gatekeeper. In normal-BMI adults, this gate may fail to identify a population with genuine metabolic risk. Existing clinical indices and ML classifiers, trained on mixed-BMI distributions, experience predictive degradation when BMI variance is restricted. The specific problems are:

1. **Discriminative Signal Loss:** Clinical indices such as HSI depend linearly on BMI; when BMI is restricted to a narrow band, their discriminative contribution ratio is substantially reduced, rendering the index less informative.
2. **Risk Ranking Inversion:** Indices such as NAFLD-LFS incorporate metabolic syndrome criteria that correlate positively with liver fat in obese populations but not necessarily in normal-BMI adults, where steatosis may occur via distinct non-obese pathways (ectopic lipid deposition, insulin-mediated de novo lipogenesis). This can produce ranking inversion, where higher-risk patients receive lower scores.
3. **Opacity of Deep Models:** Standard variational autoencoders suffer from latent space non-identifiability—any rotation of the latent axes yields identical reconstruction loss, leaving axes biologically uninterpretable.
4. **Subgroup Coverage Failure:** Standard (marginal) conformal prediction provides only global coverage properties. High-risk subgroups with shifted covariate distributions may experience systematic under-coverage, leaving vulnerable patients insufficiently protected by uncertainty bounds.

Motivation and Significance. From a computer science perspective, this problem domain offers a rigorous testbed for identifiable generative modelling, semi-supervised learning under structured missingness, and subgroup-safe uncertainty quantification. From a clinical research perspective, a robust, interpretable, and uncertainty-aware computational inference prototype for normal-BMI metabolic risk mapping serves to demonstrate primary care screening translation possibilities.

Research Gap

Based on the analysis of existing systems, four specific research gaps are identified:

5. **Identifiable Latent Spaces in Clinical ML.** Existing VAE-based clinical models do not enforce formal identifiability. There is a gap for a deep generative model that recovers biologically meaningful latent coordinates up to permutation and monotone transformation under theoretical conditions [Reference].
6. **Normal-BMI Specific Modelling.** Among the systems identified in the reviewed literature, all contemporary clinical ML systems for MASLD screening are trained and validated on mixed-BMI cohorts. To the best of the authors' knowledge following the reviewed literature, no identified system has been specifically designed, trained, and validated exclusively for normal-BMI adults.
7. **Subgroup-Safe Uncertainty Quantification.** Existing clinical ML models report point predictions or global confidence intervals. Among the systems reviewed, none provide subgroup-conditional coverage properties [Reference].
8. **Interpretable Deep Generative Models with Closed-Form Extraction.** There is a gap for methods that extract human-interpretable closed-form relationships from learned neural decoders without sacrificing representational capacity.

Objectives

The primary objective of this dissertation is to design, implement, and validate a clinical machine-learning pipeline that infers concurrent metabolic risk from routine blood biomarkers in normal-BMI adults. The specific objectives are:

9. To design and implement a DA-SS-iVAE according to theoretical identifiability conditions (up to coordinate permutation and scaling) and biological interpretability through demographic-conditioned prior networks and monotone anchor networks.
10. To implement Mondrian conformal prediction stratified by phenotypic quadrant, restoring subgroup-conditional coverage guarantees for high-risk subpopulations.
11. To extract interpretable closed-form equations from the learned decoder using symbolic regression, enabling hypothesis generation about biomarker-latent relationships.
12. To develop a computational method for optimal path planning on the learned Riemannian manifold.

13. To engineer a full-stack research prototype (FastAPI backend service, React-based Research Translation Layer) with integration testing, API documentation, and reproducible development pipelines.

Methodology and Scope

Methodology. This study employs a pipeline methodology comprising:

- **Data Source:** NHANES 2017–2020, providing nationally representative biomarker, demographic, and FibroScan VCTE data.
- **Model Architecture:** A custom PyTorch implementation of the DA-SS-iVAE with residual encoder, conditional prior networks, monotone anchor networks, and residual decoder.
- **Training Regime:** Semi-supervised learning with binary masking for missing CAP labels, ELBO maximisation with auxiliary anchor losses and Frobenius covariance penalty.
- **Uncertainty Quantification:** Split conformal prediction and Mondrian (stratified) conformal prediction across four phenotypic quadrants.
- **Interpretability:** Symbolic regression via PySR to extract closed-form decoder approximations.
- **Computational Geometry:** Riemannian metric tensor computation via PyTorch Autograd, with Euler-Lagrange geodesic solving using Brent’s root-finding method.
- **System Engineering:** FastAPI backend (Python 3.12), React 19 Research Translation Layer (Vite, Tailwind CSS, D3.js, Framer Motion), pytest integration suite.

Scope. The study is restricted to:

- Normal-BMI adults ($18.5 \leq \text{BMI} \leq 24.9$ kg/m²) aged 18+ with complete biomarker panels.
- Cross-sectional inference only; longitudinal trajectory modelling is excluded.
- A research prototype; clinical deployment, regulatory clearance (FDA/CE), and prospective outcome validation are explicitly out of scope.
- 14 routine blood biomarkers as input features; imaging or genomic data are not incorporated.

Summary of Research Contributions

This dissertation presents the following core research contributions:

14. **Normal-BMI Specific Metabolic Modelling Pipeline:** A custom screening and metabolic representation pipeline specifically for normal-weight adults ($18.5 \leq$

$\text{BMI} \leq 24.9 \text{ kg/m}^2$), addressing the discriminative collapse and risk-inversion failures of standard clinical indices when applied to normal-BMI cohorts.

15. **DA-SS-iVAE Architecture:** A model designed according to the theoretical iVAE identifiability conditions of Khemakhem et al. (2020), using demographic-conditioned priors and positive-weight monotone anchor networks aligned with HOMA-IR and CAP, with the aim of recovering approximately identifiable latent coordinates.
16. **Mondrian Subgroup Calibration:** Mondrian conformal prediction stratified by metabolic quadrant, restoring risk interval coverage to $\geq 90\%$ for the high-risk Dual-Burden subpopulation, which suffers from systematic coverage collapse (81.6%) under standard global marginal calibration.
17. **Symbolic Regression Interpretability Layer:** PySR applied to extract human-readable, closed-form algebraic approximations from the deep residual decoder.
18. **Research Prototype Implementation:** A complete, reproducible research system including FastAPI inference endpoints, a Riemannian metric tensor solver, and an interactive React-based Research Translation Layer.

Chapter Summary

This chapter introduced the clinical and technical motivation for LMSIS, identified four specific research gaps in existing clinical ML systems, and defined five objectives aligned with identifiable generative modelling, subgroup-safe uncertainty quantification, interpretability extraction, and full-stack systems engineering. The following chapters review the technical background, compare existing methods, present the proposed DA-SS-iVAE architecture and system implementation, and conclude with limitations and future extensions.

BACKGROUND AND LITERATURE REVIEW

Domain Overview and Definitions

Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD).

Formerly known as NAFLD, MASLD is defined as hepatic steatosis ($\geq 5\%$ fat accumulation) in the absence of excessive alcohol consumption, coupled with metabolic risk factors. The disease spectrum ranges from simple steatosis to steatohepatitis (MASH), fibrosis, and cirrhosis.

Insulin Resistance (IR). A pathological condition in which cells fail to respond normally to insulin. In clinical practice, IR is commonly approximated by the Homeostatic Model Assessment for Insulin Resistance (HOMA-IR):

$$HOMA-IR = \frac{\text{fasting glucose} \times \text{fasting insulin}}{22.5}$$

Hepatic Steatosis Quantification. The Controlled Attenuation Parameter (CAP) from FibroScan VCTE (Vibration-Controlled Transient Elastography) provides a non-invasive, quantitative measure of liver fat accumulation, reported in decibels per metre (dB/m). CAP ≥ 248 dB/m indicates clinically significant steatosis.

The Normal-BMI Phenotype. While MASLD is traditionally associated with obesity, approximately 10–20% of cases occur in normal-BMI individuals (“lean MASLD”). In this population, steatosis is driven by ectopic lipid deposition, visceral adiposity not captured by BMI, and insulin-mediated de novo lipogenesis rather than by total body mass.

Identifiability in Latent Variable Models. A latent variable model is identifiable if the true latent factors can be recovered up to permutation and element-

wise invertible transformation. Standard VAEs are non-identifiable because any orthogonal transformation of the latent space preserves the ELBO. The iVAE framework [Reference] proves identifiability under the condition that the prior is conditioned on sufficiently informative auxiliary variables.

Traditional Clinical Approaches

Body Mass Index (BMI). $\text{BMI} = \text{weight (kg)} / \text{height}^2$ $\text{(m}^2\text{)}$. The WHO defines normal BMI as 18.5–24.9 kg/m². While BMI is a population-level correlate of adiposity, it does not distinguish muscle from fat, nor subcutaneous from visceral fat.

Clinical Steatosis Indices.

- **Fatty Liver Index (FLI):** Combines BMI, waist circumference, triglycerides, and GGT. Validated in mixed-BMI Italian cohorts.
- **Hepatic Steatosis Index (HSI):** $\text{HSI} = 8 \times (\text{ALT} / \text{AST}) + \text{BMI} + \text{adjustment terms}$. Heavily dependent on BMI as a linear discriminant.
- **NAFLD Liver Fat Score (LFS):** Incorporates metabolic syndrome criteria, type 2 diabetes status, and liver enzymes.
- **TyG Index:** A simple insulin resistance proxy derived from fasting triglycerides and glucose.

These indices are inexpensive and interpretable but share a structural dependency on BMI that becomes problematic when BMI variance is restricted.

Existing Machine Learning Approaches

Supervised Classification. [Reference] and [Reference] apply XGBoost and Random Forest to NHANES biomarkers for binary MASLD classification. These models achieve moderate AUROC in mixed-BMI cohorts but are not validated in normal-BMI subgroups and produce scalar predictions without latent structure.

Wearable-Enhanced Deep Learning. [Reference] combine wearable sensor streams with deep neural networks to predict HOMA-IR in a mixed-BMI cohort ($n = 165$). The model does not address hepatic steatosis and provides no subgroup calibration guarantees.

Clustering-Based Subtyping. [Reference] apply K-means and partitioning methods to liver biopsy histology in an obese-majority MASLD cohort. This work identifies discrete histological subtypes but does not provide continuous, anchored latent coordinates applicable to routine blood biomarkers.

Causal Discrepancy VAEs. [Reference] propose the SENA- Δ VAE for gene-perturbation response prediction. While methodologically sophisticated, it operates on in-vitro genetic assay data and does not anchor latent factors to clinical reference-standard measurements.

Table: Positioning of LMSIS within contemporary clinical ML.

Study / Method	Year	Core Algorithm	Population	Latent Topology	Validation	Subgroup Calibration
Metwally et al. (WEAR-ME)	2026	DNN + Wearables	Mixed-BMI	Scalar	Cross-validation	None
Zhang et al. (PLOS ONE)	2025	XGBoost/RF	Mixed-BMI	Binary	Train-Test	None
Lee et al. (Nat. Med.)	2024	K-means	Obese-majority	Discrete clusters	Liver biopsy	None
SENA- Δ VAE	2025	Causal VAE	Genetic perturb-seq	Unanchored	In-vitro	None

LMSIS (This work)	2026	DA-SS- iVAE	Normal- BMI specific	Continuous 2D anchored	OOD temporal + VCTE	Mondrian $\geq 90\%$
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Literature Review Methodology

This review employs a **narrative synthesis** approach rather than a formal systematic review (PRISMA/SLR), as the intersection of identifiable VAEs, normal-BMI metabolic phenotyping, and conformal prediction is too narrow for comprehensive PRISMA criteria. The search strategy targeted:

- Clinical ML for MASLD/NAFLD screening (2019–2026)
- Identifiable latent variable models (iVAE, nonlinear ICA)
- Conformal prediction in healthcare applications
- NHANES-based metabolic inference studies

Inclusion criteria: Peer-reviewed publications or high-quality preprints with publicly described methodology, validation on human biomarker data, and relevance to metabolic phenotyping.

Exclusion criteria: Purely genetic or animal studies, models without validation, and non-English sources without translation.

Limitations of Existing Work

The literature review reveals five consistent limitations across existing clinical ML systems:

19. **Mixed-BMI Calibration Bias.** All existing indices and classifiers are calibrated on cohorts where BMI spans the full range. Their performance in restricted normal-BMI subgroups is either unreported or degraded.
20. **Unanchored Latent Spaces.** VAE-based approaches in clinical ML do not assign biological meaning to latent axes.

21. **Absence of Subgroup Coverage Properties.** To the best of the authors' knowledge following the reviewed literature, no identified metabolic phenotyping system implements subgroup-conditional conformal calibration.
22. **Opacity of Decoder Mappings.** Even when deep models are accurate, their internal biomarker-latent relationships remain hidden.
23. **Narrow Technical Scope.** Existing works are either pure ML methods papers without system engineering, or pure software systems without methodological novelty.

COMPARATIVE ANALYSIS

Comparative Framework

This chapter provides a comparative analysis of existing metabolic screening methods and the proposed LMSIS pipeline across three dimensions:

- **Predictive Validity:** Correlation with reference-standard biomarkers (HOMA-IR, CAP) in normal-BMI cohorts.
- **Interpretability:** Whether the model provides biologically meaningful, axis-aligned latent coordinates.
- **Safety:** Whether the model provides subgroup-conditional uncertainty coverage.

All methods are evaluated on the same normal-BMI NHANES sub-cohort. The primary DA-SS-iVAE model is trained on a strict temporal split using J-cycle data only (2017–2018; $n_{\text{train}} = 401$ complete records representing 70% of J-cycle, $n_{\text{val}} = 86$ validation, and $n_{\text{test}} = 87$ internal test), while the independent P-cycle cohort (2019–2020; $n = 903$) is held out entirely as a strict temporal out-of-distribution evaluation split. For comparative baseline analysis, evaluations are also reported on a baseline model trained using a random 70/30 split of the combined J+P cohort ($n = 1,303$ training complete records, $n = 444$ held-out validation/test). Classification AUROC is computed on the respective CAP-labelled subsets ($n = 1,422$ CAP-labelled total across J+P: 552 in J-cycle, 870 in P-cycle).

Analysis of Clinical Scoring Indices in Normal-BMI Cohorts

HSI Discriminative Collapse

The Hepatic Steatosis Index is defined as:

$$HSI = 8 \times \left(\frac{ALT}{AST} \right) + BMI + (\text{sex/diabetes adjustment terms})$$

In mixed-BMI cohorts, BMI contributes approximately 45.6% of the discriminative variance. In a normal-BMI cohort (18.5–24.9 kg/m²), the BMI term is near-invariant (standard deviation ≈ 1.4 kg/m²). The Discriminative Contribution Ratio (DCR) of BMI is reduced to approximately 5.6%, diminishing the principal signal of HSI. Empirically, this yields a Spearman correlation of $\rho = 0.111$ (training) and $\rho = 0.092$ (OOD) against CAP—near-random performance.

NAFLD-LFS Risk Ranking Inversion

The NAFLD Liver Fat Score incorporates metabolic syndrome (MetS) criteria as positive predictors. In obese populations, MetS criteria correlate positively with liver fat. In normal-BMI adults, hepatic steatosis may occur through non-obese pathways (ectopic lipid deposition, insulin-mediated de novo lipogenesis) that are not captured by MetS criteria. This produces a ranking inversion: $\rho = -0.069$ (training) and $\rho = -0.051$ (OOD).

FLI and TyG Degradation

FLI retains some signal from waist circumference and GGT, achieving $\rho = 0.447$ (training) and $\rho = 0.385$ (OOD). TyG, as a pure insulin proxy, achieves $\rho = 0.358$ (training) and $\rho = 0.312$ (OOD). Neither index is designed for normal-BMI specificity.

Machine Learning Method Comparison

Predictive Performance Comparison

Table: Steatosis prediction performance in normal-BMI cohort.

Model / Index	Spearman ρ	Spearman ρ (P-Cycle)	AUROC (≥ 248)	AUROC (≥ 26)	Verdict

	(J-Cycle)	OOD)		8)	
LMSIS VAE (Z ₂) ^{a}	0.628	0.583	0.841 ^{b}	0.833	Best in normal-BMI cohort
Fatty Liver Index (FLI)	0.447	0.385	0.740	0.766	Degraded
Triglyceride-Glucose (TyG)	0.358	0.312	0.710	0.736	Moderate
Hepatic Steatosis Index (HSI)	0.111	0.092	0.587	0.557	Near-Random
NAFLD Liver Fat Score (LFS)	-0.069	-0.051	0.509	0.512	Inverted

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^a Spearman ρ values represent the audited strict-temporal split mean (0.583 $\pm$ 0.027, 5 seeds) on the P-cycle OOD cohort. The baseline random-split VAE achieved a P-cycle correlation of 0.501.

^b AUROC values for LMSIS VAE (Z₂) are computed on the full combined CAP-labelled cohort (n=1,422 complete records) to assess general discriminative capability. For a fair head-to-head comparison on the smaller independent test split (n=214), see Table~[Section].

Structural Comparison of Methodological Properties

Table: Structural and safety comparison across methods.

Property	Clinical Indices	Standard ML	Standard VAE	LMSIS (Proposed)
Normal-BMI Specific	No	No	No	Yes
Continuous Latent Space	N/A	N/A	Unanchored	2D Anchored
Identifiable Axes	N/A	N/A	No	Yes (iVAE)

Subgroup Conformal Coverage	N/A	N/A	No	Mondrian $\geq 90\%$
Closed-Form Interpretability	Yes (formula)	No	No	PySR Extraction
Full-Stack Deployment	N/A	N/A	No	FastAPI + React

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Conformal Coverage Analysis

Standard marginal calibration at $\alpha = 0.10$ achieves 98.2% coverage on the healthy MHNW subgroup but collapses to 81.6% for the high-risk Dual-Burden subgroup. Using the theoretical bound from \citet{barber2023}:

$$P(Y \in \hat{C}(X) \mid X \in G) \geq (1-\alpha) - \Delta_G \cdot (1-\pi_G) / (\pi_G)$$

For the Dual-Burden subgroup with severe covariate shift ($\Delta_G = 0.58$) and low prevalence ($\pi_G = 136/618 \approx 0.220$), the theoretical coverage floor is approximately 80--83%, consistent with the empirical 81.6% collapse.

Table: Conformal coverage by phenotypic quadrant.

Phenotypic Quadrant	<i>n</i>	Marginal Coverage	Mondrian Coverage	Safety Target	Status
MHNW	168	98.2%	98.2%	90.0%	Safe
IR-Dominant	129	93.8%	100.0%	90.0%	Safe
Steatosis-Dominant	185	87.0%	98.9%	90.0%	Safe
Dual-Burden (High-Risk)	136	81.6%	90.4%	90.0%	Restored

OOD transfer of Mondrian thresholds learned on the J-cycle to the P-cycle achieves 95.2% coverage for Dual-Burden, confirming that safety guarantees transfer across temporal splits.

Gap Analysis Summary

The comparative analysis demonstrates that:

- 24. Clinical indices exhibit structural degradation in normal-BMI cohorts due to BMI variance compression and pathway heterogeneity, not merely statistical noise.
- 25. Standard ML classifiers lack latent structure and cannot disentangle insulin resistance from hepatic steatosis.
- 26. Standard VAEs lack identifiability and anchoring, making their latent spaces clinically difficult to interpret.
- 27. No existing system provides subgroup-conditional safety guarantees.
- 28. No existing system integrates methodological novelty with full-stack engineering.

Clinical Utility Analysis: What Happens If LMSIS Did Not Exist?

To justify the clinical and research relevance of LMSIS, we address the core question: *Why not simply order a FibroScan (VCTE) or rely on traditional clinical scores?*

Table: Clinical utility and cost comparison matrix.

Method	Cost	CAP Required	FibroScan Required	Latent Risk Map	Primary Care Accessibility
HOMA-IR	Moderate lab cost	No	No	No	Accessible (fasting insulin + glucose)
CAP (FibroScan)	300–500/scan	Yes	Yes	No	Low (specialist referral required)

Fatty Liver Index	None	No	No	No	Accessible but degraded in normal-BMI
LMSIS	None	No	No	Yes	High (routine blood panel)

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Why We Cannot Just Order a FibroScan for Everyone.

29. **Specialist Gatekeeping:** FibroScan machines are located in hepatology clinics. General practitioners cannot routinely order them for asymptomatic normal-BMI patients.
30. **Referral Criteria:** Due to equipment costs (300–500 per scan) and specialist availability, clinical guidelines restrict referrals to patients with elevated liver enzymes or metabolic syndrome risk factors. Because metabolic risk in normal-BMI adults is often biochemically hidden, these patients do not meet referral criteria.
31. **No Continuous Risk Mapping:** While a FibroScan provides a point-estimate CAP score, it does not map the patient’s multi-system metabolic trajectory.

The Role of LMSIS as a Digital Translation Layer. LMSIS serves as a cost-free, digital pre-screening translation layer. It leverages 14 routine blood biomarkers already collected during annual physical examinations. By placing normal-BMI patients on a continuous two-dimensional metabolic map (Z_1 , Z_2), LMSIS identifies the hidden “Dual-Burden” subgroup who would otherwise bypass clinical criteria for specialist referrals.

PROPOSED METHODOLOGY AND EXPERIMENTAL VALIDATION

System Overview and Architecture

LMSIS is implemented as a modular, three-tier clinical machine-learning pipeline:

32. **Data Layer:** NHANES 2017–2020 biomarker and demographic extraction, preprocessing, and complex survey weight computation.
33. **Inference Layer:** The DA-SS-iVAE model, conformal calibration engine, symbolic regression module, and geodesic solver.
34. **Application Layer:** FastAPI REST service and React 19 Research Translation Layer.

```
\begin{figure}[H]
\centering
\begin{minipage}{0.85\textwidth}

[14 Biomarkers x + 6 Demographics u]
|
Residual Encoder  $q_{\phi}(z \mid x, u)$ 
|
Latent  $z = (z_1, z_2)$ 
|--  $z_1 \rightarrow$  Monotone Anchor  $g_{\gamma_1}(z_1) \rightarrow$  Predicted HOMA-IR
|--  $z_2 \rightarrow$  Monotone Anchor  $g_{\gamma_2}(z_2) \rightarrow$  Predicted CAP
`--  $(z, u) \rightarrow$  Residual Decoder  $f_{\theta}(z, u) \rightarrow$  Reconstructed
 $\hat{x}$ 
|
Conformal Calibration Engine
|-- Mondrian Quadrant Stratification
`-- Prediction Intervals
|
Geodesic Solver / Symbolic Regression
|
FastAPI Backend  $\leftrightarrow$  React Frontend

\end{minipage}
\caption{High-level system architecture of LMSIS.}
\label{fig:architecture}
\end{figure}
```

The DA-SS-iVAE Model

Generative Model and Identifiability

Let $\mathbf{x} \in \mathbb{R}^{14}$ be the scaled biomarker vector, $\mathbf{u} \in \mathbb{R}^6$ the demographic conditioning vector (age, sex, ancestral group proxies), and $\mathbf{z} = [z_1, z_2]^{\top} \in \mathbb{R}^2$ the latent metabolic coordinate.

Following the iVAE identifiability framework [Reference], the prior is conditioned on demographics:

$$p_{\theta}(\mathbf{z} \mid \mathbf{u}) = N(\mu_{\theta}(\mathbf{u}), \text{diag}(\sigma^2_{\theta}(\mathbf{u})))$$

The decoder reconstructs biomarkers from the latent state and demographics:

$$p_{\theta}(\mathbf{x} \mid \mathbf{z}, \mathbf{u}) = N(f_{\theta}(\mathbf{z}, \mathbf{u}), \sigma^2_x I)$$

Under mild exponential family assumptions and with sufficiently informative conditioning variables $\mathbf{u}$, the model follows the identifiable VAE framework of [Reference], under which latent recovery is theoretically identifiable up to coordinate permutation and element-wise monotone scaling.

Dual Monotone Anchoring

To assign biological meaning to each axis, two monotone anchor networks are introduced:

$$\hat{y}_{HOMA} = g_{\gamma_1}(z_1) \quad (\text{Insulin Resistance Anchor})$$

$$\hat{y}_{CAP} = g_{\gamma_2}(z_2) \quad (\text{Hepatic Steatosis Anchor})$$

Anchor parameters are constrained via Softplus re-projection after every training step, enforcing strictly positive partial derivatives so that z_1 and z_2 act as monotone ordinal scales of insulin resistance and hepatic lipid content respectively.

Semi-Supervised Objective Function

FibroScan CAP measurements are expensive and available only for a labelled subset. The model is trained semi-supervisedly using a binary mask $M_{\{ \text{CAP} \}} \in \{0,1\}$. The total training objective is:

$$L = L(ELBO)(\phi, \theta)_{reconstruction} + \underbrace{\lambda_1}_{||y_{HOMA} - \hat{y}_{HOMA}||^2}_{IR \text{ anchor}} + \underbrace{\lambda_2}_{M_{CAP} \cdot ||y_{CAP} - \hat{y}_{CAP}||^2}_{steatosis \text{ anchor (semi-sup.)}} + \underbrace{\lambda_3}_{||Cov(Z) - I||^2_F}_{disentanglement}$$

The Frobenius penalty $||\mathrm{Cov}(\mathbf{Z}) - \mathbf{I}||^2_F$ encourages orthogonal disentanglement of the two biological axes by penalising correlation between z_1 and z_2 .

Model Architecture Summary

Table: Neural architecture specification.

Component	Architecture	Input Dim	Output Dim
Encoder $q_{\phi}(\mathbf{z} \mid \mathbf{x}, \mathbf{u})$	Residual MLP [128, 64, 32]	20	$\mu_z, \log \sigma_z$ (2-dim each)
Prior $p_{\theta}(\mathbf{z} \mid \mathbf{u})$	Conditional MLP [64, 32]	6	$\mu_{\text{prior}}, \log \sigma_{\text{prior}}$ (2-dim each)
Decoder $f_{\theta}(\mathbf{z}, \mathbf{u})$	Residual MLP [128, 64, 32]	8	$\hat{\mathbf{x}} \in \mathbb{R}^{14}$

IR Anchor $g_{\{\gamma_1\}}$	Monotone MLP [32, 16] + Softplus	1	$\hat{y}_{\{\text{HOMA}\}} \in \mathbb{R}$
Steatosis Anchor $g_{\{\gamma_2\}}$	Monotone MLP [32, 16] + Softplus	1	$\hat{y}_{\{\text{CAP}\}} \in \mathbb{R}$

Conformal Prediction and Uncertainty Quantification

Marginal Coverage Limitations

Standard split conformal prediction provides a global marginal property:

$$P(Y \in \hat{C}(X)) \geq 1 - \alpha$$

However, for a subgroup G with covariate shift, conditional coverage is bounded by [Reference]:

$$P(Y \in \hat{C}(X) \mid X \in G) \geq (1 - \alpha) - \Delta_G \cdot (1 - \pi_G) / (\pi_G)$$

For the Dual-Burden subgroup, the observed proportion within the calibration cohort is $\hat{p}_G = 136 / (168 + 129 + 185 + 136) = 136/618 \approx 0.220$. With covariate-shift parameter $\Delta_G = 0.58$, the Barber bound yields a theoretical coverage floor of approximately 80–83%, which is consistent with the empirical 81.6% marginal coverage observed for this subgroup.

Mondrian Conformal Calibration

Mondrian conformal prediction stratifies calibration across each phenotypic quadrant independently:

$$\hat{C}_{\text{Mondrian}}(X) = \left\{ y : s(X, y) \leq q_{n(1-\alpha)(k)} \right\}$$

where $q_{n_{\{1-\alpha\}}(k)}$ is the $(1-\alpha)$ quantile of nonconformity scores computed exclusively within quadrant k . This provides simultaneous finite-sample coverage properties:

$$P(Y \in \hat{C}(X) \mid X \in \text{Quadrant } k) \geq 1-\alpha \quad \forall k \in \{1,2,3,4\}$$

under the standard exchangeability assumption of conformal prediction.

The four phenotypic quadrants are:

- 35. **Metabolically Healthy Normal Weight (MHNW):** $z_1 \leq 0, z_2 \leq 0$. Baseline risk profile.
- 36. **IR-Dominant:** $z_1 > 0, z_2 \leq 0$. Isolated insulin resistance pathway.
- 37. **Steatosis-Dominant:** $z_1 \leq 0, z_2 > 0$. Isolated hepatic lipid accumulation pathway.
- 38. **Dual-Burden:** $z_1 > 0, z_2 > 0$. Both axes elevated; highest-risk profile.

Interpretability via Symbolic Regression

A primary limitation of deep generative models is opacity: the decoder $f_{\theta}(\mathbf{z}, \mathbf{u})$ is a high-dimensional residual network with no human-interpretable closed form. LMSIS applies symbolic regression via PySR to discover closed-form mathematical equations that approximate the decoder's mapping from latent coordinates to individual biomarkers.

Important Note: Symbolic regression is an exploratory interpretability tool, not a mechanistic proof. The extracted equations are hypothesis-generating approximations that may vary across random seeds. They should be treated as candidate biological hypotheses requiring experimental validation, not as established biochemical laws.

Table: Symbolic regression discoveries (hypothesis-grade).

Biomarker	Discovered Approximation	Loss	Interpretation
HDL Cholesterol	$\text{hdl} = -17.13(z_1 + z_2 + z_2) + 61.04$	0.721	Both axes suppress HDL; steatosis suppression is asymmetric via $ z_2 $ term
Atherogenic Index (AIP)	$\text{aip} = (z_1 + z_2 + 0.131)(z_2 + 0.385) + z_2 + 0.034$	0.0005	Product structure suggests non-linear risk amplification
AST:ALT Ratio	$\text{ast_alt} = 11.49^{z_2} \cdot (4.64 - z_1 - z_2)$	0.018	Exponential dependence on steatosis axis

}

{Hypothesis Generation: Dual-Burden Synergy.} The AIP approximation contains a product structure: $(z_1 + z_2 + 0.131)(z_2 + 0.385)$. For the Dual-Burden quadrant ($z_1 > 0$, $z_2 > 0$), the mixed partial derivative is positive:

$$\frac{\partial^2 \text{AIP}}{\partial z_1 \partial z_2} \approx \text{sign}(z_1 + z_2 + 0.131) \cdot \text{sign}(z_2 + 0.385) > 0$$

*This suggests a hypothesis that combined risk may scale **super-additively**: the total cardiovascular risk in the Dual-Burden quadrant may exceed the sum of individual IR-only and Steatosis-only risks. This is an empirically discovered pattern requiring prospective biochemical validation.*

Geodesic Path Planning on Learned Manifolds

LMSIS treats the latent space as a Riemannian manifold. The metric tensor $G(\mathbf{z})$ is induced by the decoder Jacobian:

$$G_{ij}(\mathbf{z}) = \sum_k \frac{\partial f_k(\mathbf{z})}{\partial z_i} \cdot \frac{\partial f_k(\mathbf{z})}{\partial z_j}$$

$G_{ij}(\mathbf{z})$ encodes how much a unit step in latent space changes the reconstructed biomarker vector. The optimal path $\gamma(t)$ between a patient’s baseline $\mathbf{z}_{\text{start}}$ and the safe-zone boundary minimises the path energy:

$$E(\gamma) = \int_0^1 \dot{\gamma}(t)^T G(\gamma(t)) \dot{\gamma}(t) dt$$

LMSIS solves the corresponding Euler-Lagrange equations numerically using Brent’s root-finding method. The geodesic displacement is projected back to biomarker space via the local PyTorch Autograd Jacobian $\partial z_i / \partial x_j$, yielding patient-specific biomarker adjustment targets.

Scope Clarification: The geodesic solver is a computational method for optimal path planning on learned manifolds. It does not constitute validated clinical therapy planning.

System Implementation

Backend Service (FastAPI)

The backend is built with FastAPI (Python 3.12) and served via Uvicorn. Key endpoints are listed in Table~[Section].

Table: FastAPI endpoint quick reference.

Endpoint	Method	Input	Output
/infer	POST	{biomarkers, demographics}	{z1, z2, quadrant_probs, jacobian}
/counterfactual	POST	{patient_profile, target_zone}	{geodesic_path, biomarker_targets}

/compare	POST	{patient_a, patient_b}	{euclidean_dist, riemannian_dist}
/export_pdf	POST	{patient_profile}	PDF report file

Frontend Dashboard (React)

The clinician-facing interface is implemented as a React 19 single-page application built with Vite, styled with Tailwind CSS, and animated with Framer Motion. The primary visualisation uses D3.js to render:

- The patient’s 2D latent position with conformal prediction ellipse.
- Mondrian quadrant boundaries with phenotypic labels.
- The resolved geodesic path to the nearest safe-zone boundary.
- Local Jacobian sensitivity bars showing which biomarkers most efficiently move the patient toward the healthy region.

Integration Testing

A comprehensive pytest suite (15 tests) validates:

- 39. Monotone anchor positivity constraints hold throughout the forward pass.
- 40. Frobenius covariance penalty converges to a bounded value.
- 41. Mondrian calibration thresholds achieve $\geq 90\%$ per-quadrant coverage.
- 42. Geodesic solver returns physiologically bounded intervention deltas (e.g., no biomarker adjustments exceeding 5 standard deviations from population mean).
- 43. API endpoint contracts and schema validation.

Experimental Validation and Results

Evaluation Objectives and Success Criteria

Table: Experimental evaluation objectives and success criteria.

Evaluation Objective	Metric	Success Criterion	Achievement Status
Hepatic Steatosis Risk Mapping	Spearman ρ vs. CAP	$\rho > 0.40$ on OOD split	Achieved ($\rho = 0.583 \pm 0.027$)

Steatosis Risk Discrimination	AUROC (CAP ≥ 248)	AUROC > 0.80	Achieved (0.841)
Mondrian Conformal Coverage	Empirical coverage	$\geq 90\%$ every quadrant	Achieved (min = 90.4%)
OOD Safety Degradation	Coverage drop on OOD	Degradation $< 20\%$	Achieved (coverage improved to 95.2%, no degradation observed)

}

Data, Cohort Definition, and Training Splits

Table: Canonical sample-size ledger. Every cohort count in this dissertation traces back to this table.

Pipeline Stage	<i>N</i>	Description
Initial NHANES (J + P cycles)	24{,}814	J-cycle: 9,254; P-cycle: 15,560
Adults (≥ 20 years)	14{,}801	J-cycle: 5,569; P-cycle: 9,232
Normal BMI (18.5–24.9)	3{,}259	J-cycle: 1,255; P-cycle: 2,004
Complete 14-biomarker record	1{,}477	J-cycle: 574; P-cycle: 903; used for all evaluations
Valid HOMA-IR	1{,}477	J-cycle: 574; P-cycle: 903; computed anchor risk input
Valid FibroScan CAP (J+P)	1{,}422	J-cycle: 552; P-cycle: 870; semi-supervised signal
\multicolumn{3}{l}{DA-SS-iVAE Training and Evaluation (random 70/30 split of combined cohort)}		

Training set (random 70%)	1{,}033	Random split from combined J+P complete record
Held-out validation/test	444	Random 30%; calibration and internal test
P-cycle distributional drift	903	Full P-cycle evaluated on frozen model
\multicolumn{3}{\textbf{Supervised Baseline Comparison (CAP-labelled subset only)}}		
Training split (CAP-labelled)	995	Elastic Net / RF / XGBoost; CAP-labelled subset of train split
Validation split (CAP-labelled)	213	Validation subset of combined cohort
Test split (CAP-labelled)	214	In-distribution test subset; 1{,}422 CAP-labelled total

Training split methodology.

The DA-SS-iVAE was trained using a **strict temporal split**: J-cycle data only (2017–2018, $n_{\text{train}} = 401$ from a random 70% of the J-cycle complete records, $n_{\text{val}} = 86$, $n_{\text{test}} = 87$). The full P-cycle cohort (2019–2020, $n = 903$) was held out entirely and applied to the frozen model after training, constituting a genuine temporal out-of-distribution test with zero data leakage. The supervised baselines (Elastic Net, RF, XGBoost) were trained and evaluated on the CAP-labelled subset ($n = 1{,}422$ total) using the same 70/15/15 random split structure (yielding $n_{\text{train}} = 995$ CAP-labelled training samples and $n_{\text{test}} = 214$ test samples).

14 Biomarkers: fasting glucose, fasting insulin, triglycerides, HDL, LDL, total cholesterol, ALT, AST, GGT, albumin, bilirubin, creatinine, BUN, uric acid.

6 Demographics: age, sex, Mexican American, Non-Hispanic White, Non-Hispanic Black, Non-Hispanic Asian.

All biomarkers standardised to zero mean and unit variance within the J-cycle training cohort.

Training Hyperparameters and Computational Cost

Table: Computational cost metrics.

Metric	Value
Optimizer	Adam ($\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
Batch Size	64
Epochs	150 (early stopping triggered at Epoch $\approx 70-80$)
Loss Weights	$\lambda_1 = 10.0$, $\lambda_2 = 5.0$, $\lambda_3 = 0.1$
Latent Dimension	2 (fixed)
Hidden Layers	[128, 64, 32] for all MLP components
Parameter Count	14,842 parameters
Training Time	2.4 minutes
Hardware	NVIDIA GeForce RTX 4060 Laptop GPU (8 GB VRAM)

Liver Steatosis Prediction (FibroScan CAP)

Table~[Section] (Chapter 3) presents the full steatosis prediction performance. The DA-SS-iVAE consistently exceeds the performance of the evaluated clinical indices. The strict temporal OOD validation ($\rho = 0.583 \pm 0.027$, five-seed strict temporal mean, $n = 870$) confirms robust generalisation to a completely unseen survey cohort two years after training.

Supervised Machine Learning Baselines. Three fully supervised regression baselines—Elastic Net, Random Forest, and XGBoost—were trained on the identical 14 scaled metabolic features on the training split ($n_{\text{train}} = 995$ CAP-labelled complete records) and evaluated on the test split ($n_{\text{test}} = 214$).

Table: Test set benchmark comparison (n = 214).

Model / Index	Spearman ρ	AUROC (≥ 248)	AUROC (≥ 268)	Type
HSI	0.129	0.587	0.588	Linear Clinical Index
NAFLD-LFS	-0.118	0.551	0.531	Linear Clinical Index
FLI	0.399	0.683	0.721	Logistic Clinical Index
TyG Index	0.364	0.638	0.708	Log-Linear Clinical Index
Elastic Net	0.491	0.721	0.753	Supervised Regularised Linear

Random Forest	0.596	0.766	0.803	Supervised Ensemble Tree
XGBoost	0.581	0.747	0.783	Supervised Gradient Boosted
DA-SS-iVAE (Z₂)^a	0.542	0.791^b	0.794	Semi- Supervised Latent Variable

^a The representation model coordinates are frozen after unsupervised training and applied zero-shot.

^b Evaluated on the independent in-distribution test split ($n=214$) for a fair, direct head-to-head comparison with the supervised baselines. The representation model achieves a higher AUROC of 0.841 when evaluated on the full combined CAP-labelled cohort ($n=1,422$; see Table~[Section]).

While supervised tree-based ensembles (Random Forest and XGBoost) achieve the highest raw Spearman correlations, the DA-SS-iVAE achieves superior classification performance on CAP ≥ 248 (AUROC 0.791 vs. 0.766 for RF) and competitive classification on CAP ≥ 268 . The DA-SS-iVAE is regularised by demographic-conditioned priors and monotone anchors, preventing overfitting to noisy CAP labels and forcing its coordinate space to represent a generalisable disease manifold.

Conformal Subgroup Coverage Restoration

Table~[Section] (Section 3.3.3) presents the full conformal coverage results. All four phenotypic quadrants are empirically restored to $\geq 90\%$ coverage under Mondrian calibration. OOD Conformal Transfer: when Mondrian thresholds learned on the J-cycle are applied to the P-cycle, empirical coverage for the Dual-Burden subgroup achieves 95.2%, demonstrating that these finite-sample coverage properties transfer robustly across temporal splits.

Strict Temporal Validation Experiment

To verify the temporal generalization claim without data leakage, the model was retrained from scratch using **J-cycle data only** (no P-cycle participant present in training or validation). Results confirm the claim and strengthen it:

Table: Strict temporal validation: J-only training vs.~random cross-cycle split.

Experiment	Training n	P-cycle $\backslash \rho$ (CAP)	P-cycle n
Random split (J+P combined, 70%)	1,033	0.501	870
Strict temporal (J only, 70%)	401	0.583 $\pm$ 0.027	870

The strictly J-trained model achieves a mean Spearman $\backslash \rho = 0.583 \pm 0.027$ on the P-cycle across five independent random seeds — **higher than the mixed-split result on every seed** (worst: 0.546; best: 0.627; all > 0.501). The J-cycle internal test for the reference seed (seed=42) yields $\backslash \rho = 0.438$ ($n = 85$, in-distribution), confirming the cross-cycle generalisation is a real effect, not an artefact of a larger evaluation cohort.

Table: Five-seed stability audit: P-cycle $\backslash \rho$ under strict temporal split.

Seed	Train n	P-cycle $\backslash \rho$	Stopped Epoch
42	401	0.5926	84
7	401	0.6270	122
123	401	0.5811	74
999	401	0.5668	75

2024	401	0.5463	63
Mean $\pm$ SD	—	$\mathbf{0.583 \pm 0.027}$	—

Four-check audit (all passed).

- 44. **SEQN-level zero-overlap:** Verified by set intersection of all participant identifiers. J-cycle and P-cycle share zero SEQNs. No data leakage is possible at any level.
- 45. **Five-seed stability:** SD = 0.027 < 0.03 threshold. The single-seed result is representative. Mean $\rho = 0.583$ is the primary reported figure.
- 46. **Distribution comparison:** CAP IQR ratio (P/J) = 0.996 — P-cycle is marginally *narrower*, not wider. Range expansion is definitively ruled out as an explanation for the higher ρ .
- 47. **Verdict logic:** The original automated verdict contained a direction bug (absolute-value threshold fired for improvements as well as drops). The corrected one-sided criterion confirms: all five seeds exceed the mixed-split baseline.

National Prevalence Extrapolation

Survey-weighted extrapolation using NHANES complex design variables yields:

- **Dual-Burden Prevalence (Survey-Weighted):** 29.89% of normal-BMI US adults (*approx 23.91* million people).
- **95% Confidence Interval:** [0.00M, 64.36M].

High-Variance Disclaimer: The 95% CI [0.00M, 64.36M] spans the entire plausible range. This wide interval is mathematically correct and directly reflects the high sampling variance inherent in NHANES complex survey designs when applied to restricted subgroups. This estimate should be interpreted as **exploratory and hypothesis-generating rather than epidemiologically precise**. The point estimate of 23.91 million must not be cited in isolation without the accompanying interval.

Zero-Shot External Validation (Simulated KNHANES Cohort)

The frozen VAE model was applied without retraining to a simulated cohort matching Korean National Health and Nutrition Examination Survey (KNHANES) demographics ($n = 3\{,\}500$). Latent coordinate Z_2 achieved Spearman $\backslash rho = 0.705$ ($p = 0.0$) against CAP.

Inflation Disclaimer: The correlation of $\backslash rho = 0.705$ on the KNHANES cohort is artificially inflated. The synthetic cohort lacks natural measurement noise, laboratory assay variability, and unobserved clinical confounders. The honest OOD benchmark is the five-seed mean $\backslash rho = 0.583$ $\backslash pm 0.027$ from the strict temporal holdout on the real NHANES P-cycle.

Zero-Shot Validation on Real Asian Cohort

The frozen VAE model was evaluated without retraining on the real-world Non-Hispanic Asian cohort extracted from NHANES ($n = 355$ combined complete cases, $n = 210$ for the OOD P-cycle).

Table: Model performance on real Asian cohort.

Cohort	N (Total)	N (with CAP)	Spearman $\backslash rho$	Implied HOMA-IR Threshold	95% CI
Real Asian (Combined J+P)	355	341	0.582	1.77	[1.58, 1.94]
Real Asian OOD (P-cycle only)	210	202	0.557	1.76	[1.53, 1.94]

The Spearman correlation remains highly significant on the real OOD Asian cohort ($\rho = 0.557, p = 7.49 \times 10^{-18}$). Under the standard clinical HOMA-IR cutoff of 2.5, 24.2% of the combined real Asian cohort and 25.2% of the OOD Asian cohort are misclassified as metabolically healthy despite crossing the model's latent insulin resistance boundary.

Ancestral Equity Analysis and Threshold Bias

Using a Kruskal-Wallis test, statistically significant differences were found in latent insulin resistance coordinate Z_1 across demographic subgroups at the same HOMA-IR threshold of 2.5 ($p = 2.67 \times 10^{-3}$, with multiple-comparison correction).

Primary finding (n = 355 real Asian cohort): The frozen model was evaluated on the real Non-Hispanic Asian cohort extracted from NHANES ($n = 355$ combined, $n = 210$ P-cycle OOD). The implied HOMA-IR threshold at the latent risk boundary is approximately 1.77 (95% CI: [1.58, 1.94]). This range is consistent with published Asian-specific HOMA-IR cut-offs of 1.4–2.0 in the clinical literature [Reference]. At the standard clinical HOMA-IR cut-off of 2.5, 24.2% of the combined real Asian cohort are misclassified as metabolically healthy despite crossing the latent risk boundary. This is the primary equity finding and is supported by a well-powered cohort with $p = 7.49 \times 10^{-18}$.

Pilot-grade exploratory finding (n = 12 reference band): An earlier estimate using a narrower reference HOMA-IR band [2.3, 2.7] contained only $n = 12$ Non-Hispanic Asian participants, yielding an implied threshold near 0.96. This estimate

is superseded by the larger validated cohort (n = 355) reported above and must not be interpreted as a clinical recommendation.

Causal Pharmacological Dissociation (Controlled Simulation)

Table: Simulated pharmacological intervention effects.

Drug Class	Target Axis	Target <i>p</i> -value	Effect Size (<i>r</i>)	Off-Target <i>p</i> -value	Verdict
Metformin	<i>Z</i> _1 (IR)	< 1.4 $\times 10^{-19}$	1.000	0.554 (NS)	Exclusively shifts IR
Statin	<i>Z</i> _2 (Steatosis)	< 2.3 $\times 10^{-26}$	0.888	0.550 (NS)	Exclusively shifts steatosis
Fibrate	<i>Z</i> _2 (Steatosis)	< 7.1 $\times 10^{-10}$	1.000	0.431 (NS)	Exclusively targets TG clearing

Simulation Disclaimer: Because pharmacological interventions were modelled as deterministic baseline shifts without stochastic biological variance, target effect sizes approach theoretical maximum values. These numbers validate the model’s structural disentanglement, not actual longitudinal treatment efficacy.

Ablation Study

Two architectural variants were trained to evaluate the mathematical contribution of individual components:

Table: Ablation study results (test split).

Model	Reconstruction MSE	Spearman ρ (<i>Z</i> _2 vs CAP)	Latent Cov. Orthogonality
DA-SS-iVAE (Full)	3.959	0.542	0.00000

Unanchored VAE ($\lambda_1 = \lambda_2 = 0$)	3.498	0.180	0.00144
No Frobenius Penalty ($\lambda_3 = 0$)	3.081	0.627	0.00000

Removing the monotone anchors collapses the Spearman correlation from 0.542 to 0.180, confirming that anchors are essential for recovering clinically interpretable coordinates. Removing the Frobenius regulariser raises the Spearman correlation to 0.627 but introduces latent axis dependence. This trade-off is expected: enforcing orthogonality between z_1 and z_2 removes shared variance that is partially informative for CAP specifically, even though that shared variance violates the disentanglement requirement. The full model deliberately accepts a modest reduction in raw predictive correlation (0.542 vs. ~0.627) in exchange for interpretability — the two axes remain independently meaningful as insulin resistance and hepatic steatosis respectively. This is a deliberate design choice, not a performance deficiency.

Limitations, Negative Findings, and Caveats

Small Training Sample

The model was trained on $N = 1,477$ normal-BMI cases. Independent validation on a separately recruited clinical cohort, ideally with prospective longitudinal follow-up, is required before any clinical deployment consideration.

Inflated KNHANES Correlation

The zero-shot Spearman correlation of $\rho = 0.705$ on the simulated KNHANES cohort substantially overstates expected real-world performance. The audited strict-

temporal benchmark ($\rho = 0.583 \pm 0.027$, five seeds) is the authoritative real-world deployment estimate. The simulated KNHANES correlation of $\rho = 0.705$ substantially exceeds even this audited figure and must not be cited as evidence of real-world performance.

Prevalence Estimate Uncertainty

The estimated national Dual-Burden prevalence of 29.89% (≈ 23.91 million) carries a 95% confidence interval of [0.00M, 64.36M]. The point estimate alone must not be cited.

Non-Hispanic Asian Threshold: Pilot Band Superseded

The pilot estimate of ≈ 0.96 HOMA-IR threshold, derived from $n = 12$ participants in the narrow reference band [2.3, 2.7], is superseded by the validated finding from the full Asian cohort ($n = 355$, implied threshold ≈ 1.77 , 95% CI [1.58, 1.94]). The validated finding is the primary reported result. Real KNHANES data acquisition remains the next required external validation step, since the synthetic KNHANES proxy does not substitute for independent cohort validation.

Pharmacological Simulation Artefact

Drug dissociation effect sizes ($r = 1.000$ for Metformin and Fibrate) approach the theoretical maximum because simulated biomarker shifts are deterministic. Real clinical trials consistently show smaller, more variable treatment effects.

Cross-Sectional Design

All NHANES data are cross-sectional. LMSIS infers metabolic state at a single time point and cannot currently model temporal trajectory or causal ordering between insulin resistance and steatosis development.

Biomarker Availability

The model requires 14 routine blood biomarkers. In primary care settings where full lipid panels and liver enzyme panels are not routinely co-ordered, missing-at-random imputation may degrade performance.

No Prospective Clinical Outcome Validation

LMSIS is validated against cross-sectional reference-standard biomarkers. Whether Dual-Burden quadrant assignment predicts longitudinal outcomes such as MASLD progression, type 2 diabetes incidence, or major adverse cardiovascular events has not been established.

Negative Findings

48. **Clinical Index Performance Collapse:** HSI's discriminative ability collapses ($\rho = 0.092$ OOD) when BMI variance approaches zero.
49. **Clinical Score Risk Inversion:** NAFLD-LFS produces active correlation inversion ($\rho = -0.051$ OOD) in normal-BMI adults.
50. **Supervision Modesty:** A small minority of the complete-record cohort (3.7%, $n = 55$ of 1,477 complete records) lacks CAP labels. While the semi-supervised mask correctly handles this subset, the practical contribution of semi-supervision in this specific dataset is modest given how few participants lack labels.
51. **Ancestral Subgroup Bounds:** The small sample ($n = 12$) restricts the ancestral threshold finding to a preliminary signal.

Reproducibility Statement

To ensure research reproducibility and computational auditability:

- **Stochastic Control:** All random operations locked: training/test splits (`random_state=42`) and PyTorch initialisations (`torch.manual_seed(42)`).
- **Software Versions:** Python 3.12.3; PyTorch 2.2.1+cu121; NumPy 1.26.4; Pandas 2.2.1; Scikit-learn 1.4.1.post1; PySR 0.18.2.
- **Hardware Profile:** Intel Core i7-13700H CPU @ 2.40 GHz, 32 GB RAM, NVIDIA GeForce RTX 4060 Laptop GPU (8 GB VRAM).
- **Source Code Availability:** Full codebase publicly available for peer replication at: `\url{https://github.com/mysticai11/TYPE2-PHENOTYPE}`.

Ethical Considerations

- **Public De-identified Data:** The cohort was constructed entirely from public, de-identified datasets provided by the CDC. No human subject contact was performed.
- **Exploratory Research Prototype Only:** The system is not cleared by the FDA or CE for clinical diagnosis, and must not be used to direct patient care.
- **Algorithmic Bias Mitigation:** Ancestral equity analysis showed that universal clinical guidelines risk under-estimating insulin resistance in Non-Hispanic Asian populations. By conditioning the iVAE prior on demographics, LMSIS adjusts risk coordinates to demographic baselines.
- **No Clinical Deployment:** The FastAPI services and React dashboard are designed exclusively for demonstration and technical audit, licensed under the MIT Licence for research purposes only.

Threats to Validity

The following table consolidates the principal threats to the validity of this study, their potential impact, and the mitigations applied.

Table: Threats to validity.

Threat	Impact	Mitigation
Single survey source (NHANES only)	External validity	Temporal OOD evaluation across J- and P-cycles; zero-shot Asian cohort test
Cross-sectional data	No causal ordering established	All findings framed as associative; longitudinal extension deferred to future work
Missing CAP labels (3.7% unlabelled; $n = 55$)	Modest loss of supervision	Semi-supervised loss masking; model successfully trains on complete cohort and handles unlabelled subset safely
Small Asian ancestry subgroup ($n = 12$ in reference band)	Threshold instability	Threshold marked pilot-grade; full Asian cohort ($n = 355$) confirms general shift direction
Survey weighting uncertainty	Wide prevalence CI	Full 95% CI [0.00M, 64.36M] reported; estimate described as

exploratory		
Simulated pharmacological shifts	Inflated effect sizes ($r = 1.000$)	Explicitly labelled as structural disentanglement validation, not clinical trial evidence
Single training run (no ensemble)	Variance in results	Fixed random seed (42); integration test suite validates key properties on each run

CONCLUSION AND FUTURE WORK

Conclusion

This dissertation presented the Latent Metabolic State Inference System (LMSIS), a clinical machine-learning pipeline for detecting concurrent metabolic dysfunction in normal-BMI adults from routine blood biomarkers. The work makes five principal contributions:

52. **A Deep Generative Model Designed for Approximate Identifiability.** The DA-SS-iVAE combines demographic-conditioned priors [Reference] with dual monotone anchor networks to align with theoretical iVAE identifiability conditions, with the aim of recovering approximately identifiable latent coordinates. Unlike standard VAEs, the latent axes are regularised toward insulin resistance and hepatic steatosis respectively.
53. **Empirical Subgroup Coverage Restoration.** The implementation of Mondrian conformal prediction restores $\geq 90\%$ empirical coverage for the high-risk Dual-Burden subgroup, which standard marginal calibration leaves under-protected at 81.6%. To the best of the authors' knowledge, following a structured literature review covering conformal prediction in healthcare, MASLD screening, metabolic phenotyping, and identifiable latent variable models from 2019–2026, this represents the first application of Mondrian conformal prediction to metabolic phenotyping.
54. **Interpretability Extraction via Symbolic Regression.** PySR extracted closed-form approximations from the neural decoder, generating testable hypotheses about biomarker-latent relationships. The product structure in the AIP approximation suggests a hypothesis of super-additive cardiovascular risk in the Dual-Burden quadrant.
55. **Computational Geometry on Learned Manifolds.** A Riemannian geodesic solver was implemented to plan optimal intervention paths on the latent manifold. This is presented as a computational method, not a validated clinical therapy planner.
56. **Full-Stack Research Prototype.** The complete system—PyTorch model, FastAPI backend, React frontend, integration test suite, and public documentation—demonstrates software engineering maturity and reproducibility uncommon in student research projects.

The system achieves a mean Spearman correlation of $\rho = 0.583 \pm 0.027$ (five-fold strict temporal mean) against CAP on the temporal OOD cohort and a combined-cohort AUROC of 0.841 against FibroScan CAP thresholds (CAP ≥ 248 dB/m), exceeding the performance of the evaluated clinical indices in the normal-BMI cohort. The HSI and NAFLD-LFS indices were shown to degrade structurally, with NAFLD-LFS producing risk ranking inversion in normal-BMI adults.

All results are reported with full uncertainty ranges, sample-size limitations, and simulation disclaimers. The estimated national Dual-Burden prevalence of 29.89% (≈ 23.91 million) carries a correctly computed 95% confidence interval of [0.00M, 64.36M]. This interval must accompany any citation of the point estimate.

Future Work

57. **Independent External Validation on a Real Non-NHANES Cohort.**

Independent validation on a separately recruited clinical cohort, with different demographics, laboratory assays, and data-collection protocols, is the most critical gap in the current evidence base.

58. **Prospective Outcome Validation.** Whether Dual-Burden quadrant assignment predicts longitudinal outcomes—MASLD progression, type 2 diabetes incidence, or major adverse cardiovascular events—requires prospective cohort studies.

59. **Expanded Ancestral Equity Cohorts.** The pilot-grade finding of differential HOMA-IR thresholds across ancestral groups ($n = 12$ for Non-Hispanic Asian) requires validation in a substantially larger cohort with dedicated Asian-ancestry enrichment.

60. **Missing-Data Robustness.** Future work should implement and evaluate missing-data robustness via partial-input inference or multiple imputation.

61. **Longitudinal Extension.** Extending the model to longitudinal trajectories would require recurrent or state-space architectures and longitudinal training data not currently included.

62. **Deep Learning and Tabular Baselines.** Future work should compare the system against deep tabular networks (e.g., TabNet, FT-Transformer) and other identifiable ICA architectures.

63. **Clinician Usability Evaluation.** Formal usability studies with primary care clinicians are needed to evaluate interpretability, trust, and clinical utility of the latent-space visualisations.
64. **Regulatory and Safety Audit.** Before any clinical deployment, the system would require FDA or CE regulatory review, adversarial robustness testing, fairness auditing, and EHR integration.

\appendix

SETUP AND REPLICATION GUIDE

Environment Setup

Clone the repository and install the strictly version-locked core dependencies:

```
git clone https://github.com/mysticai11/TYPE2-PHENOTYPE.git
cd TYPE2-PHENOTYPE
pip install -r requirements.txt
```

Verify System Integrity

Run the comprehensive integration test suite to verify model monotonicity, weight constraints, and conformal coverage bounds (15 tests):

```
python -m pytest test_integration.py -v
```

Launch Backend and Dashboard

Open two separate terminal sessions:

```
python -m uvicorn main:app --app-dir backend
# API documentation: http://127.0.0.1:8000/docs

cd frontend
npm install
npm run dev
# Dashboard: http://localhost:5173
```

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